

CHILDREN ADHD DISEASE DETECTION USING POSE ESTIMATION TECHNIQUES

ABSTRACT

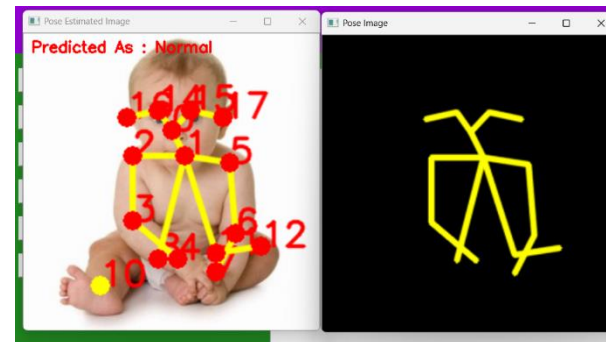
Attention Deficit Hyperactivity Disorder (ADHD) is a common neurodevelopmental disorder in children characterized by inattention, hyperactivity, and impulsive behaviour. Early detection is essential for effective treatment and academic support. However, traditional diagnostic methods rely on behavioural observation and clinical evaluation, which can be subjective and time-consuming.

This project proposes an automated ADHD detection system using pose estimation techniques. By analysing children's movement patterns, posture stability, and activity levels from video data, the system identifies behavioural indicators associated with ADHD. Using computer vision and machine learning models, the proposed solution provides a non-invasive, objective, and efficient method for early screening and assessment.

DESCRIPTION

Modern healthcare increasingly depends on AI-based diagnostic tools, yet ADHD diagnosis still largely relies on manual observation, behavioral questionnaires, and psychological assessments, which can be subjective and time-consuming. To address these limitations, we developed an AI-based system that analyzes children's movement patterns using pose estimation techniques. The system captures video data during various activities and extracts body key points through advanced pose estimation algorithms. It then evaluates motion intensity, posture variations, and movement frequency to detect signs of hyperactivity and impulsive behavior. These extracted features are processed using machine learning models to classify behavioral traits associated with ADHD. By providing objective, data-driven insights, the system reduces subjectivity in diagnosis and supports healthcare professionals in early screening. It is scalable, cost-effective, and suitable for implementation in schools and clinical environments.

Attention Deficit Hyperactivity Disorder (ADHD)



EXISTING SOLUTION

1. Clinical Interviews based on DSM guidelines
2. Parent and Teacher Rating Scales
3. Neuropsychological Tests
4. Classroom Behavioral Observation
5. Wearable Activity Monitoring Devices

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CONCLUSION

The proposed ADHD detection system leverages pose estimation and machine learning to automate behavioural analysis in children. By providing objective and early screening, the system supports healthcare professionals in accurate diagnosis and timely intervention.

This AI-driven approach improves accessibility, reduces diagnostic delays, and enhances decision-making in child mental healthcare. With further development and validation, it can become a valuable tool in educational and clinical environments.

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